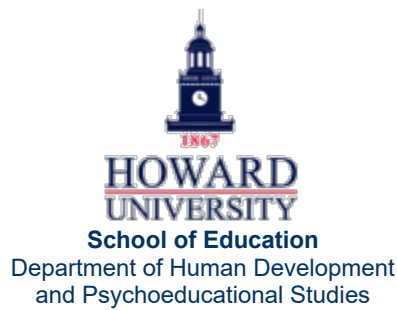




## HUDE 400-01: INTERMEDIATE STATISTICS

### SPRING 2026

**Course:** HUDE 400  
**Room:** Miner Room 212  
**Instructor:** Kyndra V. Middleton, Ph.D.  
**Email:** [kyndra.middleton@howard.edu](mailto:kyndra.middleton@howard.edu)  
**Office Hours:** Tuesday: 2:30 – 5:30PM (in-person) | Wednesday: 8 – 9AM (hybrid) | Friday: noon – 1PM (virtual) | Other times: by appointment (send an email)

**Credit Hours:** 3  
**Meeting Time:** Tuesday 7:40-10PM  
**Office:** Miner Room 201X  
**Phone:** (202) 806-5342

**Office Hours Note:** I have a distinct awareness of the value of time. As such, there is no need to make an appointment during my regularly scheduled office hours. I will be in the office during the posted times unless I have notified the class beforehand. Email verification is not necessary.

#### **Course Description**

This course provides an introduction to quantitative reasoning in empirical research, particularly the execution and interpretation of commonly used descriptive and inferential statistical techniques. The principles learned in this course provide a foundation for a general understanding of quantitative social science methodology. If you ever want to collect data, analyze data, critically read an article, which presents a data analysis, or think about the relationship between theory and the real world, then you will find this class useful. Overall, the objectives are to develop your skills in modeling and carrying out your own research, and to prepare you for more advanced graduate courses in social science statistics.

**Required Readings:** Any handouts given during class or via Canvas

**Required Technology:** A scientific or graphing calculator as well as access to SPSS

#### **Course Requirements**

**Assignments and Grading:** The following assessments are used to evaluate your mastery of the topics.

##### **Labs (200 points)**

There will be eight labs throughout the course. All labs must be completed on or before the due date (no later than 11:59PM on the Sunday following the lecture on that topic [beginning on 1/25/26]). Any lab submissions after the due date will receive a 5-point deduction each day the assignment is not submitted (beginning at 8AM Monday).

##### **Application Assignments (300 points)**

There will be a total of two application assignments due this semester. You may work individually or in groups to complete these assignments. All application assignments must be uploaded to Canvas on or before the due date (no later than 11:59PM on Sunday, 2/22/26 and Sunday, 4/19/26). Any application assignment submissions

after the due date will receive a 5-point deduction each day the assignment is not submitted (beginning at 8AM Monday).

### **Project (300 points)**

There will be a project due in its entirety on April 21<sup>st</sup> in which you must incorporate material learned in class to examine a research topic. Components of the project will be due throughout the semester so everyone is consistently working toward the end goal.

### **Tests (200 points)**

Two tests will be given during the semester. Tests will be cumulative and will be constructed based on material covered, homework problems, and any handouts provided. Tests must be taken on the date scheduled unless an extenuating circumstance exists where you must take the test early or late (within one week of the original test date). If the test is taken late, it must be completed no later than 7:40PM on the following Tuesday.

### **Competencies Addressed**

2A3. Demonstrates basic knowledge and understanding of research design and statistical analyses and applies appropriate research methods and data analysis techniques to address identified problems

4A3. Demonstrates understanding of the basic psychometric constructs (e.g., validity, reliability)

### **Learning Objectives**

Students will

1. Competency: Demonstrate basic knowledge and understanding of research design and statistical analyses and applies appropriate research methods and data analysis techniques to address identified problems.  
Measure: lab work, application assignments, project, and cumulative tests
2. Competency: Demonstrate understanding of the basic psychometric constructs (e.g. validity, reliability)  
Measure: lab work, application assignments, project, and cumulative tests
3. Competency: Demonstrate the ability to apply concepts of Intermediate Statistics  
Measure: lab work, application assignments, project, and cumulative tests
4. Competency: Demonstrate the ability to analyze data  
Measure: lab work, application assignments, project, and cumulative tests
5. Competency: Demonstrate the ability to identify the appropriate design for their research study  
Measure: lab work, application assignments, project, and cumulative tests

### **Course Rubric**

| Learning Objectives Addressed              | Demonstration Level 1                       | Demonstration Level 2  | Demonstration Level 3         | Demonstration Level 4     |
|--------------------------------------------|---------------------------------------------|------------------------|-------------------------------|---------------------------|
| Students will demonstrate understanding of | No understanding of Intermediate Statistics | Basic understanding of | Intermediate understanding of | Advanced understanding of |

|                                                                                                   |                                                                        |                                                                           |                                                                                  |                                                                              |
|---------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------------|----------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| Intermediate Statistics                                                                           |                                                                        | Intermediate Statistics                                                   | Intermediate Statistics                                                          | Intermediate Statistics                                                      |
| Student will demonstrate the ability to apply concepts of Intermediate Statistics                 | No ability to apply concepts of Intermediate Statistics                | Basic ability to apply concepts of Intermediate Statistics                | Intermediate ability to apply concepts of Intermediate Statistics                | Advanced ability to apply concepts of Intermediate Statistics                |
| Students will demonstrate the ability to analyze data                                             | No knowledge and skill necessary to analyze data                       | Basic knowledge and skill necessary to analyze data                       | Intermediate knowledge and skill necessary to analyze data                       | Advanced knowledge and skill necessary to analyze data                       |
| Students will demonstrate the ability to identify the appropriate design for their research study | No ability to identify the appropriate design for their research study | Basic ability to identify the appropriate design for their research study | Intermediate ability to identify the appropriate design for their research study | Advanced ability to identify the appropriate design for their research study |

## Grading

Final course grades will be assigned as follows:

| Assignment   | Measurement                        | Expected Result                                     |
|--------------|------------------------------------|-----------------------------------------------------|
| Labs         | 20%<br>8 Lab Assignments           | 90% of students will receive a grade of B or higher |
| Applications | 30%<br>Two Application Assignments | 90% of students will receive a grade of B or higher |
| Project      | 30%<br>Project Deliverables        | 95% of students will receive a grade of B or higher |
| Tests        | 20%<br>Two Exams                   | 85% of students will receive a grade of B or higher |
| Total        | 100%                               |                                                     |

Final course letter grades will be assigned as follows:

| Letter Grade | Percent of Grade | Point Range |
|--------------|------------------|-------------|
| A            | 90-100%          | 900-1000    |
| B            | 80-89%           | 800-899     |
| C            | 70-79%           | 700-799     |
| D            | 60-69%           | 600-699     |

**Learner Expectations:**

- (1) You are expected to attend every class, as you are responsible for the material covered. If, however, you miss a class, it is your responsibility to find out what was discussed and make up any missed work.
- (2) You are expected to complete and submit all assignments by the due date. Any assignment submitted after 11:59PM on the due date will receive an automatic 5-point deduction. Each additional day the application assignment or lab is late, there will be an additional 5-point deduction.

**Tips for Success:** You will find the following bit of advice quite useful. (1) Don't get behind in the material. Be sure to study the material before and after the material is presented in lecture. (2) Read the material before it is covered in class. It is possible you might need to read the material several times. (3) Seek help immediately when you do not understand a concept or want extra practice. (4) Review your notes and complete homework assignments immediately after the material is covered. (5) Keep your notes and homework organized for easy reference. (6) Actively participate in class.

**Additional Policies**

**Class Attendance Restricted to Registered Students:** Only students whose names appear on the official course roster are permitted to attend class meetings. Students who are not registered are not permitted to attend or participate in course activities, do not have access to Canvas, cannot submit course assignments, and will not receive a grade for this course. It is the students' responsibility to ensure that they are properly registered by the published registration deadline. Requests to add courses after the deadline will not be considered.

**Participation:** Learning is a collaborative process, and this course is structured around the assumption of active collaboration and co-ownership. Although the Professor will take responsibility for the overall design and direction of the course, all participants in the course must share in creating a rich intellectual environment and a fruitful dialogue. Therefore, please come to each class and lab meeting prepared to participate actively and respectfully, to ask questions, to discuss the salient issues and problems that emerge from the readings, and to utilize and contribute your knowledge and professional experiences in addressing the course material. Active, thoughtful, respectful participation and contribution in class and lab are required in this course.

**Cell Phone Usage:** The use of cell phones, smart phones, iPods, or other mobile communication devices is disruptive and distractive and is therefore discouraged during live sessions. Anyone who wishes to use one of these devices should turn their camera off. During a quiz or an exam, the use of any of these devices is not allowed (including the calculator on the device).

**Academic Integrity:** The University is committed to principles of scholastic honesty. An act of plagiarism, defined as "Any passing off of another's ideas, words, or work as one's own") is considered to be a violation. The instructor in this course reserves the right to use computerized detection systems to help prevent plagiarism. Currently, Howard University subscribes to Turnitin.com for purposes of plagiarism screening. By enrolling in this course, students agree that all assignments are subject to submission for textual similarity review to Turnitin.com. Please note that any assignments or course documents submitted to this service will be included as source documents for the restricted access database of Turnitin.com which is exclusively used for the purpose of detecting plagiarism. Material identified as plagiarized will be dealt with pursuant to University's Student Code

of Conduct. Penalties for plagiarism can be severe, up to and including expulsion from the University. Additionally, all assignments, unless otherwise noted, must be completed independently. Any quiz or exam given during class must remain in the room unless it has been graded previously.

**Statement of Academic Honesty:** Howard University expects its students to function within an environment of trust relative to other students, faculty, staff, and administration. Moreover, the School expects all students to conduct themselves ethically, with personal honesty, and with professionalism. Academic dishonesty violates one of the most basic ethical principles in an Academic community, and will result in sanctions imposed under the School's disciplinary system. All incidents of Academic dishonesty must be immediately referred to the Department Chair or Program Director for investigation and intervention. The Executive Committee may be convened to review the student's situation. Possible interventions and sanctions may include, but are not limited to, developing a remediation plan, placing a student on Academic warning, suspending or dismissing a student. Academic dishonesty includes, but is not limited to:

*CHEATING:* In any form, including but not limited to, giving or receiving aid on tests, giving or receiving test materials prior to official distribution, or collaborating on assignments or exams without instructor permission.

*PLAGIARISM:* The use or reproduction of another's work without appropriate attribution. The School expects all students to produce original work in their papers, coursework, dissertation, and other Academic projects and to follow appropriate rules governing attribution.

*FABRICATION:* Inventing information or citations in an Academic or clinical exercise.

*Additionally, all assignments, unless otherwise noted, must be completed independently. Any quiz or exam given during class must remain in the room unless it has been graded previously. Any form of cheating will result in an automatic grade of 0.*

**Artificial Intelligence (AI) Policy:** This statement is written to provide guidance around the use and misuse of AI, specifically Generative AI. You may use AI sources, e.g., ChatGPT to help generate ideas and brainstorm. However, you should note that the material generated by these programs may be inaccurate, incomplete, or otherwise problematic. Beware that use may also stifle your own independent thinking and creativity. Thus, all work submitted in this course must be your own. Contributions from anyone or anything else, including AI sources, must be properly quoted and cited every time they are used. Failure to do so constitutes an academic integrity violation, and the institution's policy may be enacted to the letter in those instances.

**Title IX:** Howard University takes sexual assault, dating violence, domestic violence, stalking and sexual harassment seriously. If a student reveals that he or she needs assistance with any of these issues, all Responsible Employees, which includes faculty, are required to share this information with the University Title IX Office (202-806-2550). Students can be referred for confidential services to the Interpersonal Violence Prevention Program (IVPP) (202 806-7647) or University Counseling Services (202 806-6870).

**Statement of ADA Procedures:** Howard University is committed to providing an educational environment that is accessible to all students. In accordance with this policy, students in need of accommodations due to a disability should contact the Office of the Dean for Special Student Services for verification and determination of reasonable accommodations as soon as possible after admission to the University or at the beginning of each semester. The Dean of Special Student Services can be reached at (202) 283-2420.

**Religious Accommodations:**

Students who wish to observe religious holidays must inform their instructors in writing within the first two weeks of each semester/term of their intent to observe the holiday so that reasonable alternative arrangements convenient to both students and faculty can be made.

## Course Schedule

| Week Of | Student Responsibilities                        | Material Covered                                                                                                                                                          |
|---------|-------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1/13    | First Day of Class (Show Up!)                   | <ol style="list-style-type: none"> <li>1. Introductions</li> <li>2. Course Information</li> <li>3. Pretest</li> <li>4. Review</li> </ol>                                  |
| 1/20    | Become familiar with T-Tests                    | <ol style="list-style-type: none"> <li>1. T-Test</li> <li>2. T-Test Lab Due</li> </ol>                                                                                    |
| 1/27    | Become familiar with One-Way ANOVA              | <ol style="list-style-type: none"> <li>1. One-Way ANOVA</li> <li>2. One-Way ANOVA Lab Due</li> </ol>                                                                      |
| 2/3     | Become familiar with Factorial ANOVA            | <ol style="list-style-type: none"> <li>1. Factorial ANOVA</li> <li>2. Factorial ANOVA Lab Due</li> </ol>                                                                  |
| 2/10    | Become familiar with Repeated Measures ANOVA    | <ol style="list-style-type: none"> <li>1. Repeated Measures ANOVA</li> <li>2. Repeated Measures ANOVA Lab Due</li> </ol>                                                  |
| 2/17    | Application                                     | <ol style="list-style-type: none"> <li>1. Application Assignment (T-Test, One-Way, Factorial, Repeated Measures ANOVA)</li> </ol>                                         |
| 2/24    | Become familiar with Mixed ANOVA                | <ol style="list-style-type: none"> <li>1. Mixed ANOVA</li> <li>2. Mixed ANOVA Lab</li> </ol>                                                                              |
| 3/3     | Cumulative Midterm Exam                         | <ol style="list-style-type: none"> <li>1. T-Test, One-Way, Factorial, Repeated Measures, Mixed ANOVA</li> </ol>                                                           |
| 3/10    | SPRING BREAK!                                   | <b>NO CLASS (ENJOY!)</b>                                                                                                                                                  |
| 3/17    | Become familiar with Chi-Square                 | <ol style="list-style-type: none"> <li>1. Chi-Square</li> <li>2. Chi-Square Lab</li> </ol>                                                                                |
| 3/24    | Become familiar with Correlation and Regression | <ol style="list-style-type: none"> <li>1. Correlation and Regression</li> </ol>                                                                                           |
| 3/31    | Continue with Regression                        | <ol style="list-style-type: none"> <li>1. Regression</li> <li>2. Regression Lab</li> </ol>                                                                                |
| 4/7     | Become familiar with ANCOVA                     | <ol style="list-style-type: none"> <li>1. ANCOVA</li> <li>2. ANCOVA Lab</li> </ol>                                                                                        |
| 4/14    | Project Work; Application                       | <ol style="list-style-type: none"> <li>1. Finalize Projects</li> <li>2. Application Assignment (Mixed ANOVA, Chi-Square, ANCOVA, Correlation &amp; Regression)</li> </ol> |
| 4/21    | Project Presentations Review                    | <ol style="list-style-type: none"> <li>1. Presentations</li> <li>2. Project Submissions Due</li> <li>3. Final Exam Review</li> </ol>                                      |
| 4/28    | Final Exam                                      | <ol style="list-style-type: none"> <li>1. Everything covered in the course</li> </ol>                                                                                     |